

Verifying Ohm's Law

PhET Ohm's Law · PhET

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

NOTICE — AI-generated worksheet

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Simulation: <https://phet.colorado.edu/en/simulations/ohms-law>

Curriculum links: circuits · electricity

Learning intentions

- I can describe the relationship between voltage, current and resistance in a simple circuit.
- I can collect and graph data to verify Ohm's Law.
- I can calculate an unknown quantity in a circuit using $V = IR$.

Getting started

Open PhET Ohm's Law. Use the sliders to set the resistance and voltage of the simple circuit shown, and read the current from the on-screen meter or the animated flow of charge. Keep resistance constant for the first set of tasks, then keep voltage constant for the second set.

Tasks

1. Set the resistance to a fixed value. Record the resistance, then note the current reading at this setting.

2. Keeping resistance constant, vary the voltage across five different values. Record the voltage and resulting current each time.

Voltage (V)	Resistance (Ω , constant)	Current (A)

3. Using your table from Task 2, calculate $V \div I$ for each trial. What do you notice about this value across all five trials?

Working:

4. Based on Task 3, describe in words the relationship between voltage and current when resistance is held constant.

5. Now keep voltage constant and vary resistance across five different values. Record resistance and current each time.

Resistance (Ω)	Voltage (V, constant)	Current (A)

6. Describe the relationship between resistance and current when voltage is held constant, based on your second table.

7. Using Ohm's Law ($V = IR$), calculate the current you would expect for a circuit with resistance $15\ \Omega$ and voltage 9 V . Check your answer by setting up this exact circuit in the simulation.

Working:

8. A circuit has a current of 0.5 A flowing through a 20 Ω resistor. Calculate the voltage across the resistor, then verify your answer using the simulation.

Working:

Extension (optional)

A household appliance is labelled '240 V, 2 A'. Use Ohm's Law to calculate its resistance, and explain what would happen to the current if the appliance's resistance increased due to a fault, given the voltage supply stays the same.
